

EXERCISE 1

Thermodynamics of electrochemical cells

April 27, 2026 | O. Schmidt | Theory and Computation of Energy Materials (IET-3)

AIM OF TODAY'S EXERCISES

Today, we investigate the thermodynamic effects of electrochemical cells.
The following link may be helpful

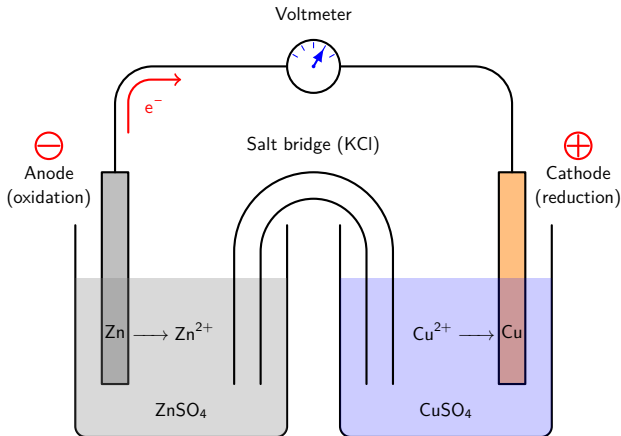
→ [Exercises](#) ←

Objectives: The student is expected to be able to devise/ derive and/ or calculate

- The cell designation, the half cell reaction, the cell reaction, and the standard cell potential.
- The cell potential at non-standard conditions.
- The electrode potential.

when provided with sufficient material.

THE GALVANIC CELL: DANIELL CELL

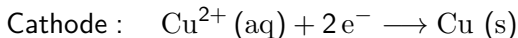
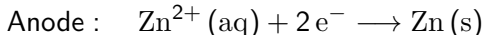


THE GALVANIC CELL: WORKED EXAMPLE

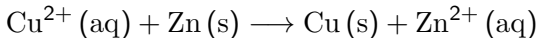
Cell designation: We have the following cell designation



Half-cell reactions: The half-cell reactions are



Cell reaction: The cell reaction is found to be

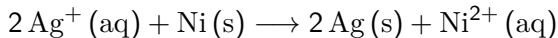


Standard cell potential: Using the Table on slide 5

$$E_{\text{cell}}^{\ominus} = E_{\text{red}}^{\ominus}(\text{Cu}^{2+}/\text{Cu}) - E_{\text{red}}^{\ominus}(\text{Zn}^{2+}/\text{Zn}) = 0.34\text{ V} - (-0.76\text{ V}) = \underline{\underline{1.10\text{ V}}}$$

THE GALVANIC CELL: QUESTIONS

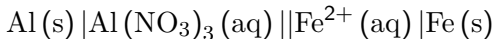
1.1) For the following electrochemical reaction



1.1a) Write the half-cell reactions.

1.1b) Calculate the standard cell potential.

1.2) Draw a diagram of the following galvanic cell with a salt bridge of your choice



1.3) On slide 6, there is a diagram. For this diagram, please

1.3a) Write the cell designation.

1.3b) Write the half-cell and cell reaction(-s).

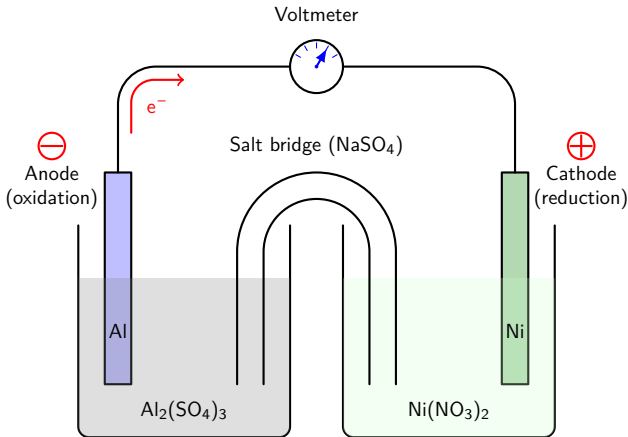
1.3c) Calculate the standard cell potential.

ELECTROCHEMICAL SERIES

Half-cell			$E_{\text{red}}^{\ominus}$ (V)
$\text{Ag}^+(\text{aq}) + \text{e}^-$	$\rightarrow$	$\text{Ag}(\text{s})$	+0.80
$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^-$	$\rightarrow$	$\text{Cu}(\text{s})$	+0.34
$\text{Ni}^{2+}(\text{aq}) + 2\text{e}^-$	$\rightarrow$	$\text{Ni}(\text{s})$	-0.25
$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^-$	$\rightarrow$	$\text{Fe}(\text{s})$	-0.44
$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^-$	$\rightarrow$	$\text{Zn}(\text{s})$	-0.76
$\text{Al}^{3+}(\text{aq}) + 3\text{e}^-$	$\rightarrow$	$\text{Al}(\text{s})$	-1.66

* Values are measured at 1 bar and 298.15 K.

THE GALVANIC CELL: QUESTION 1.3



THE CELL POTENTIAL

Nernst equation

Let E_{cell} be the cell potential and Q be the reaction quotient of an electrochemical reaction, then the cell potential is found using the Nernst equation:

$$\text{Nernst' equation : } E_{\text{cell}} = E_{\text{cell}}^{\ominus} - \frac{RT}{\nu F} \ln(Q)$$

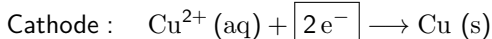
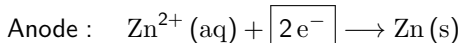
Furthermore, at equilibrium we have $Q = K$, and no potential is generated by the galvanic cell, i. e. $E_{\text{cell}} = 0$. Therefore

$$\text{Equilibrium : } E_{\text{cell}}^{\ominus} = \frac{RT}{\nu F} \ln(K) \implies K = \exp\left(\frac{\nu F E_{\text{cell}}^{\ominus}}{RT}\right)$$

THE CELL POTENTIAL: WORKED EXAMPLE

Consider the Daniell cell example from before.

Number of electrons, ν : We repeat the half-cell reactions for completion here and box the number of equal equation



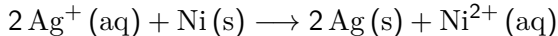
i. e. the number of electrons in the half-cell reactions is $\nu = \underline{\underline{2}}$.

The equilibrium constant: Insert the values

$$K = \exp \left(\frac{\nu F E_{\text{cell}}^{\ominus}}{RT} \right) = \exp \left(\frac{2 \times 96485 \text{ C mol}^{-1} \times 1.10 \text{ V}}{8.314 \text{ J mol}^{-1} \times 298.15 \text{ K}} \right) \approx \underline{\underline{1.55 \times 10^{37}}}$$

THE CELL POTENTIAL: QUESTIONS

2.1) For the following half cell reactions



2.1a) How many electrons participate in the half-cell reactions?

2.1b) Calculate the equilibrium constant of the reaction under standard conditions.

2.2) Consider the following cell



how much does the cell potential change, when Q is increased by a factor 10?

CHEMICAL POTENTIAL IN THE IDEAL DILUTE LIMIT

Activity: Chemical potential in the dilute limit

Let $\mu_A^{(\text{pure})}$ be the chemical potential of a mixture of pure A, a_A be the activity of compound A, μ_A be the chemical potential of compound A. Then the chemical potential is provided by

$$\mu_A = \mu_A^{(\text{pure})} + RT \ln(a_A)$$

and in the ideal-dilute limit, we have

$$\mu_A \approx \mu_A^{(\text{pure})} + RT \ln(x_A)$$

which is shown in the appendix.

APPENDIX: IDEAL LIMITS

Ideal case: Raoult's law

Consider an arbitrary compound A. Let $p_A^{(\text{pure})}$ be the vapour pressure of pure A, $p_A^{(\text{solution})}$ be the vapour pressure of A immersed in a solution, and x_A be the mole fraction. For an ideal solution, the following holds true

$$p_A^{(\text{solution})} = p_A^{(\text{pure})} x_A$$

Ideal-dilute case: Henry's law

Let K_A be an empirical constant, then the following holds true in for an ideal dilute solution

$$p_A^{(\text{solution})} = K_A x_A$$

APPENDIX: ACTIVITY

Activity and ideal solution

Consider a solvent with compound A. The activity of compound A is defined by

$$\mu_A = \mu_A^{(\text{pure})} + RT \ln(a_A)$$

and we can measure this value using the vapour pressure

$$a_A = \frac{p_A^{(\text{solution})}}{p_A^{(\text{pure})}}$$

APPENDIX: ACTIVITY + IDEAL DILUTE

Activity: Chemical potential in the dilute limit

Applying the activity definition, measurement expression, Henry's law and $\log(ab) = \log(a) + \log(b)$

$$\begin{aligned}\mu_A &= \mu_A^{(\text{pure})} + RT \ln(a_A) = \mu_A^{(\text{pure})} + RT \ln\left(\frac{p_A^{(\text{solution})}}{p_A^{(\text{pure})}}\right) \\ &= \mu_A^{(\text{pure})} + RT \ln\left(\frac{K_A x_A}{p_A^{(\text{pure})}}\right) = \mu_A^{(\text{pure})} + RT \ln\left(\frac{K_A}{p_A^{(\text{pure})}}\right) + RT \ln(x_A)\end{aligned}$$

And if Raoult's law holds, then $x_A p_A^{(\text{pure})} = K_A x_A \implies p_A^{(\text{pure})} = K_A$ and from $\log(1) = 0$, we have

$$\mu_A = \mu_A^{(\text{pure})} + RT \ln(x_A)$$